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Thermodynamic assessments of the Yb–Bi and the Yb–Te systems

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ABSTRACT

The Yb–Bi and the Yb–Te systems are thermodynamically assessed by CALPHAD approach based on the available experimental data including thermochemical properties and phase equilibria. All of the intermetallic phases in these two systems are treated as stoichiometric compounds due to their unnoticeable homogeneity ranges, and their formation enthalpies at 0 K are calculated by the first-principles method to supplement the thermochemical property data for the present optimization. The values of the first-principles calculation are confirmed by the lattice constants and the equilibrium modulus of the compounds available from experiments. Given the asymmetric shapes of the liquidus around the compounds Yb_4Bi_3/YbTe in the Yb–Bi/Te phase diagrams, both the associate model and the sub-regular solution model are used to describe the Gibbs free energy of the liquid phase for the two systems. The comparison between the optimization results and the experimental data for the Yb–Bi/Te systems shows that a much better agreement is obtained by means of the associate model than the sub-regular solution model.

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1. Introduction

Rare earth (RE) elements, such as Ce or Yb, are often added into the Bi_2Te_3-based thermoelectric materials to increase phonon-scattering, and accordingly to raise the Seeback coefficient and to improve thermoelectric properties [1–5]. Therefore the RE–Bi/Te systems deserve to be studied in order to accomplish the RE–Bi–Te thermodynamic database for developing high-performance thermoelectric materials. The phase diagrams of the related systems, especially the calculated phase diagrams using CALPHAD (CALculation of PHAse Diagrams) method, may be straightforward and convenient to offer the guidance for materials design. The thermochemical properties and the phase equilibria concerning the Yb–Bi/Te systems would be helpful for a better understanding of the interactions of Bi/Te with Yb.

As a part of the RE–Bi–Te thermodynamic database, this work presents the optimization of the thermodynamic descriptions for the Yb–Bi/Te binary systems based on the available experimental data and the present first-principles calculations.

2. Evaluation of the available information

2.1. The Yb–Bi system

The phase diagram of the Yb–Bi binary system was previously measured by Borzone [6] based on thermal analysis, micrographic and X-ray diffraction methods. The whole composition range was studied and five intermetallic compounds, Yb_2Bi, Yb_5Bi_3, Yb_4Bi_3, $Yb_{11}Bi_{10}$ and $YbBi_2$, were reported. The Yb_4Bi_3 phase melts congruently at about 1753 K, whereas the other four intermetallic compounds are formed from the peritectic reactions, $L+Yb_4Bi_3 \leftrightarrow Yb_5Bi_3$ at 1673 K, $L+Yb_5Bi_3 \leftrightarrow Yb_2Bi$ at 1373 K, $L+Yb_4Bi_3 \leftrightarrow Yb_{11}Bi_{10}$ at 1173 K, and $L+Yb_{11}Bi_{10} \leftrightarrow YbBi_2$ at 923 K. There are two eutectic reactions, $L \leftrightarrow$ Bcc_A2 (Yb) + Yb_2Bi at 1053 K and $L \leftrightarrow$ Rhombohedral_A7 (Bi) + $YbBi_2$ at 523 K. All the intermetallic compounds and the terminal phases are of very limited solubility ranges, and they are treated as the stoichiometric compounds and the pure element phases respectively in the present work. The detailed invariant reactions of the Yb–Bi system are shown in Table 1.

The thermochemical data of the Yb–Bi binary system were scarcely reported in literature. The enthalpy of mixing of the liquid phase at 50 at% Bi was calculated by Takeuchi and Inoue [7] based on Miedema model [8], which is a semi-empirical model for predicting thermochemical properties, especially the enthalpies of formation of compounds and the enthalpies of mixing of solutions. The formation enthalpy of the intermetallic compound, Yb_4Bi_3, was measured by direct calorimetry method [9] and predicted by

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Table 1
The invariant reactions of the Yb–Bi binary system.

Invariant reaction	Reaction type	T(K)	x_{Bi}^{L}	Reference
$L \leftrightarrow Yb_2Bi$ + Bcc_A2 (Yb)	Eutectic	1053	~0.05	[6]
		1056	0.03	This work^S
		1056	0.03	This work^A
$L + Yb_5Bi_3 \leftrightarrow Yb_2Bi$	Peritectic	1373	~0.16	[6]
		1373	0.17	This work^S
		1373	0.15	This work^A
$L + Yb_4Bi_3 \leftrightarrow Yb_5Bi_3$	Peritectic	1673	~0.3	[6]
		1673	0.34	This work^S
		1673	0.32	This work^A
$L \leftrightarrow Yb_4Bi_3$	Congruent	1753	0.4286	[6]
		1754	0.4286	This work^S
		1753	0.4286	This work^A
$L + Yb_4Bi_3 \leftrightarrow Yb_{11}Bi_{10}$	Peritectic	1173	~0.6	[6]
		1173	0.64	This work^S
		1173	0.63	This work^A
$L + Yb_{11}Bi_{10} \leftrightarrow YbBi_2$	Peritectic	923	~0.72	[6]
		922.5	0.71	This work^S
		923	0.73	This work^A
$L \leftrightarrow YbBi_2$ + Rhomb_A7 (Bi)	Eutectic	523	~0.95	[6]
		528	0.94	This work^S
		523	0.94	This work^A

"S" refers to the sub-regular solution model, and "A" the associate model.

Table 2
The invariant reactions of the Yb–Te binary system.

Invariant reaction	Reaction type	T(K)	x_{Te}^{L}	Reference
$L \leftrightarrow$ YbTe + Bcc_A2 (Yb)	Eutectic	1093	~0.005	[11]
		1090	0.005	[12]^S
		1093	0.003	This work^A
$L \leftrightarrow$ YbTe	Congruent	2003	0.5	[11]
		2003.46	0.5	[12]^S
		2003	0.5	This work^A
$L \leftrightarrow$ YbTe + Hexagonal_A8 (Te)	Eutectic	726	~0.993	[11]
		721.70	0.997	[12]^S
		722.32	0.998	This work^A

"S" refers to the sub-regular solution model, and "A" the associate model.

Table 3
The calculated lattice constants a, b and c compared with available experimental data.

Phase	Pearson symbol	Space group	a (Å)	b (Å)	c (Å)	Reference
Yb_5Bi_3-hex	hP16	$P6_3/mcm$	9.165	–	6.983	First-principles
			9.185	–	7.039	Experiment [31]
Yb_5Bi_3-orth	oP32	Pnma	12.639	9.705	8.337	First-principles
			12.638	9.722	8.407	Experiment [32]
Yb_4Bi_3	cI28	I43d	9.554	–	–	First-principles
			9.573	–	–	Experiment [33]
$YbBi_2$	oS12	Cmcm	4.642	16.682	4.401	First-principles
			4.56	16.68	4.28	Experiment [34]
YbTe	cF8	Fm3m	6.348	–	–	First-principles
			6.345	–	–	Experiment [35]

Cho et al. [10] also based on Miedema model.

2.2. The Yb–Te system

The phase equilibria of the Yb–Te binary system were measured previously by Abrikosovet al. [11] and optimized recently by Ghamri et al. [12]. It was reported [11] that one intermetallic compound, YbTe, and three terminal solution phases, Bcc_A2 (Yb), Fcc_A1 (Yb) and Hexagonal_A8 (Te), exist in the system. The compound YbTe melts congruently at about 2003 K. There are two eutectic reactions, $L \leftrightarrow$ Bcc_A2 (Yb) + YbTe at 1093 K and $L \leftrightarrow$ Hexagonal_A8 (Te) + YbTe at 723 K. The intermetallic compound and the terminal phases were also treated as the stoichiometric compound and the pure element phases respectively in the present work for the same reason as their very limited solubility ranges. The detailed invariant reactions of the Yb–Te system are shown in Table 2.

The enthalpy of formation of the intermetallic compound, YbTe, was measured by Spychiger et al. [13] using the fluorine combustion calorimetry [14], and predicted by Mills [15] and by Eichler et al. [16] based on Miedema model. Petzel and Ludwigs [17] also estimated the enthalpy of formation of YbTe from their studies of the thermodynamic vaporization.

3. Calculation method

3.1. First-principles calculations

In order to supplement the experiment data of the thermochemical properties for the intermetallic compounds existing in the Yb–Bi/Te systems and to obtain more reliable thermodynamic descriptions for these two systems, the first-principles method is used to calculate the enthalpies of formation of the binary compounds, Yb_2Bi, Yb_5Bi_3, Yb_4Bi_3, $Yb_{11}Bi_{10}$, $YbBi_2$ and YbTe. The scalar relativistic all-electron Blöchl's projector augmented-wave (PAW) [18,19] method within the generalized gradient approximation (GGA) is adopted, as implemented in the highly-efficient Vienna Ab initio Simulation Package (VASP) [20,21]. For the GGA exchange-correlation functional, the Perdew–Burke–Ernzerhof parameterization (PBE) [22–24] is employed. All the structures were fully relaxed using the Methfessel–Paxton method pertaining to volume and atomic coordinates. For calculations of the element involving magnetic properties, such as Yb, spin-polarized calculations were performed. For consistency, a plane-wave energy cutoff is held constant as 400 eV for the A_xB_y compounds (A = Yb, B = Bi or Te, and x and y are the stoichiometric numbers of the elements A and B respectively in the related compound), and Brillouin zone integrations are performed using Monkhorst–Pack [25,26] K-point meshes ($12 \times 12 \times 12$ in the present work). The total energy is converged numerically to less than 0.001 eV/atom.

The formation enthalpies of the intermetallic compounds are calculated by the following equation:

$$ {}_fE\left(A_xB_y\right) = E_{total}\left(A_xB_y\right) - xE_{total}(A) - yE_{total}(B) \quad (1)$$

where $E_{total}(A_xB_y)$, $E_{total}(A)$ and $E_{total}(B)$ are the calculated total energies (per atom at $T=0$ K) of the intermetallic compound A_xB_y, pure A and pure B, respectively.

Since the effect of pressure on the condensed phases is ignored and energies are calculated at 0 K without any entropic contributions, the energy of formation is considered to be the enthalpy of formation.

3.2. Thermodynamic modeling

The lattice stabilities of all pure elements are adopted from the SGTE (Scientific Group Thermodata Europe) compilation by Dinsdale [27]. Different models are employed to describe the Gibbs free energies of the liquid phase and the intermetallic compounds in the Yb–Bi/Te binary systems.

3.2.1. Liquid phase

Both the associate model and the sub-regular solution model are used to describe the Gibbs free energies of the liquid phases of the Yb–Bi/Te systems for comparison between the optimization results and the experimental data.

3.2.1.1. Sub-regular solution model. Based on random mixing of the constituent atoms, the liquid phase of the Yb–Bi/Te systems is described. The Gibbs free energy of the liquid phase is expressed with the following expression [28]:

$$G_m^{Liq} = x_A{}^0G_A^{Liq} + x_B{}^0G_B^{Liq} + RT(x_A \ln x_A + x_B \ln x_B) + x_Ax_B \sum_{i=0}^{n} {}^iL_{A,B}^{Liq}(x_A - x_B)^i \quad (2)$$

where, R and T are the gas constant and the absolute temperature, respectively; x_k and ${}^0G_k^{Liq}$ are the molar fraction and the molar Gibbs free energy of the pure component k (k=A and B, A=Yb and B=Bi or Te); and ${}^iL_{A,B}^{Liq}$ is the ith ordered interaction parameter between species A and B, which is formulated as a linear function of temperature ${}^iL_{A,B}^{Liq} = a_i + b_iT(i=0, 1, 2,\ldots)$. The coefficients a_i and b_i are the parameters to be optimized in the present work. For the Yb–Te system, the liquid can be represented by using only the ${}^0L_{A,B}^{Liq}$ and ${}^1L_{A,B}^{Liq}$ terms, and is called a subregular solution. Whereas for the Yb–Bi system, the liquid requires the ${}^0L_{A,B}^{Liq}$, ${}^1L_{A,B}^{Liq}$ and ${}^2L_{A,B}^{Liq}$ terms to get an optimized description, and is called a sub-subregular solution. In the present work, the 'sub-regular' is used unitively for simplicity.

3.2.1.2. Associate model. The short range ordering behavior exists in the liquid phase of many Bi-containing binary systems, such as Li–Bi, Na–Bi etc. Given the asymmetric shape of the liquidus boundary around the compound Yb_4Bi_3 in the Yb–Bi phase diagram, the associate model [29] is used to describe the Gibbs free energy of the liquid phase. In the meantime, for the Yb–Te system, taking into account the facts that the optimization results by Ghamri et al. [12] using sub-regular solution model is not very well in agreement with the available experimental data and that the asymmetric shape of the liquidus boundary around the compound YbTe exists in the Yb–Te phase diagram, the associate model [29] is also used to describe the Gibbs free energy of the liquid phase.

Corresponding to the compounds with the highest congruent melting points in the Yb–Bi/Te systems, Yb_4Bi_3 and YbTe are considered to be the constituent associate species besides the pure components Yb and Bi/Te in the liquid phases of the systems. Consequently, the Gibbs free energy of the liquid phase is expressed with the following expression [28]:

$$G_m^{Liq} = \sum_{i=1}^{3} x_i{}^0G_i^{Liq} + RT\sum_{i=1}^{3} x_i \ln x_i + \sum_{i=1}^{2}\sum_{j=i+1}^{3} x_ix_j \sum_{n=0}^{m} {}^nL_{i,j}^{Liq}(x_i - x_j)^n \quad (3)$$

where, ${}^0G_i^{Liq}$ and x_i are the molar Gibbs free energy and the mole fraction of the pure species i (i = 1, 2, 3 represent Bi, Yb, Yb_4Bi_3 respectively in the Yb–Bi system and Te, Yb, YbTe respectively in the Yb–Te system) in the liquid state; ${}^nL_{i,j}^{Liq}$ is the nth ordered interaction parameter (n = 0, 1, 2 in the present work) between the species i and j (j = 2, 3 represents Yb, Yb_4Bi_3 respectively in the Yb–Bi system and Yb, YbTe respectively in the Yb–Te system), which is a function of temperature as ${}^nL_{i,j}^{Liq} = a_n + b_nT$; and the coefficients a_n and b_n are the parameters to be optimized in the present work.

3.2.2. Intermetallic compounds

There are totally six intermetallic compounds in the Yb–Bi/Te binary systems. All these compounds, Yb_2Bi, Yb_5Bi_3, Yb_4Bi_3, $Yb_{11}Bi_{10}$, $YbBi_2$ and YbTe, are treated as the stoichiometric compounds A_mB_n, since no noticeable homogeneity ranges are reported for these compounds. Meanwhile, due to lack of the heat capacities of these phases, the Gibbs free energy per mole of atoms is expressed for each of the intermetallic compounds, according to the Neumann–Kopp rule [30]:

$$G_m^{A_mB_n} = \frac{m}{m+n}{}^0G_A^{SER} + \frac{n}{m+n}{}^0G_B^{SER} + c + dT \quad (4)$$

where, m and n are the stoichiometric coefficients of the

Table 4
The calculated equilibrium energy (E_0), equilibrium volume (V_0) compared with available experimental data.

Phase	E_0(eV)	V_0(Å^3)	Reference
Yb_5Bi_3-hex	−53.151	507.882	First-principles
	–	514.3 [31]	Experiment
Yb_4Bi_3	−48.669	872.079	First-principles
	–	877.29 [33]	Experiment
$YbBi_2$	−22.615	340.8	First-principles
	–	325.54 [34]	Experiment
TeYb	−8.004	255.806	First-principles
	–	255.44 [35]	Experiment

Table 5
The formation enthalpies of compounds at 0 K obtained by the first-principles calculations, the Miedima model and the reported experimental data.

Phase	Experimental (J/mol)	First-principles in this work (J/mol)	Miedima model (J/mol)
Yb_2Bi (using Sm_2Bi structure)	–	−79961.7	–
Yb_5Bi_3-hex	–	−88574.6	–
Yb_5Bi_3-orth	–	−87893.8	–
Yb_4Bi_3	−91,000 [9]	−90208.9	−65,800 [10]
$Yb_{11}Bi_{10}$ (using $Yb_{11}Sb_{10}$ structure)	–	−84147.9	–
$YbBi_2$	–	−58728.3	–
YbTe	−155326.3 [13]	−165129.3	−120,000 [16]

Table 6
The thermodynamic parameters of the Yb–Bi system by using the associate model for the liquid phase.

Phase	Thermodynamic parameter
Liquid (Bi, Yb, Yb_4Bi_3)	$G^{Liq}_{Yb_4Bi_3} = -465153 + 8.8T + 4*G_YB_LIQ + 3*G_BI_LIQ$
	$G_BI_LIQ = \begin{cases} (298.15 < T < 455.54) \\ +11246.066 - 20.636509*T + GHSERBI - 5.9549 \times 10^{-19}*T^{+7}; \\ (455.54 < T < 800) \\ +11336.26 - 20.810418*T + GHSERBI - 1.66145 \times 10^{+25}*T^{-9}; \\ (800 < T < 1200) \\ +11336.259 - 20.810418*T + GHSERBI - 1.66145 \times 10^{+25}*T^{-9}; \\ (1200 < T < 3000) \\ +3754.947 + 103.961021*T - 27.196*T*Ln(T). \end{cases}$
	$G_YB_LIQ = \begin{cases} (298.15 < T < 1033) \\ +7030.788 - 40.615571*T - 1.8061816*T*Ln(T) - 0.03250938*T^2 \\ +5.136665 \times 10^{-6}*T^3 - 370554*T^{-1}; \\ (1033 < T < 2000) \\ -6445.835 + 186.690398*T - 36.7774*T*Ln(T). \end{cases}$
	${}^0L^{Liq}_{Bi,Yb} = -220000 + 0.6634T$
	${}^1L^{Liq}_{Bi,Yb} = +2710.115-0.72T$
	${}^2L^{Liq}_{Bi,Yb} = +38000 + 5.7663T$
	${}^0L^{Liq}_{Yb,Yb_4Bi_3} = +50000$
	${}^0L^{Liq}_{Bi,Yb_4Bi_3} = -90497.9$
	${}^1L^{Liq}_{Bi,Yb_4Bi_3} = -178583-2.5T$
	${}^2L^{Liq}_{Bi,Yb_4Bi_3} = +33872.8$
Yb_2Bi	$G^{Yb_2Bi}_{Yb:Bi} = -83700.595 + 10.52T + 0.6667GHSERYB + 0.3333GHSERBI$
Yb_5Bi_3	$G^{Yb_5Bi_3}_{Yb:Bi} = -92059.5 + 10.92T + 0.6250GHSERYB + 0.3750GHSERBI$
Yb_4Bi_3	$G^{Yb_4Bi_3}_{Yb:Bi} = -93062.355 + 7.465T + 0.5714GHSERYB + 0.4286GHSERBI$
$Yb_{11}Bi_{10}$	$G^{Yb_{11}Bi_{10}}_{Yb:Bi} = -87185.649 + 5.93T + 0.5238GHSERYB + 0.4762GHSERBI$
$YbBi_2$	$G^{YbBi_2}_{Yb:Bi} = -61330.498 + 2.59T + 0.3333GHSERYB + 0.6667GHSERBI$

compound A_mB_n; ${}^0G^{SER}_A$ and ${}^0G^{SER}_B$ are the molar Gibbs free energies for the pure components A and B in their Standard Element Reference (SER) states, Face Centered Cubic (FCC_A1) for Yb, Rhombohedral_A7 for Bi and Hexagonal_A8 for Te. The parameters *c* and *d* are to be optimized in the present work.

4. Results and discussion

4.1. First-principles calculations

The available experimental lattice parameters are used as the starting points to perform the structural calculations for the compounds, Yb_5Bi_3, Yb_4Bi_3, $YbBi_2$ and YbTe [31–35], while for the other two compounds, $Yb_{11}Bi_{10}$ and Yb_2Bi, no experimental lattice data are reported. The total energies for the different ratios of the lattice constants *c*/*a* and *b*/*a* are optimized around the equilibrium cell and the obtained values of these two equilibrium parameters are fixed when calculating the equilibrium volume using Birch–Murnaghan equation of state. In Table 3 the optimized results of the lattice constants *a*, *b* and *c* are summarized and compared with the available experimental results taken from Refs. [31–35]. The present values of the optimized lattice parameters are roughly 1% higher than the experimental results. The data of the equilibrium volume (V_0) and the equilibrium energy (E_0) are listed in Table 4.

Based on Eq. (1), the enthalpies of formation of the binary compounds of the Yb–Bi/Te systems are calculated. The comparisons between the calculated results and the experimental data as well as the Miedema model's predictions are listed in Table 5. The values of the first-principles calculations for Yb_4Bi_3 and YbTe are consistent with those reported by Ferro et al. [9] and by Ghamri et al. [12] with the relative deviations within $\pm 0.87\%$ and $\pm 6.3\%$ respectively, resulting in an error bar on the calculated enthalpy of formation of ± 2 kJ/mol in the Yb–Bi system and ± 8 kJ/mol in the Yb–Te system.

Compared with the available experimental data, including the lattice constants and the equilibrium energies of all compounds as well as the reported enthalpies of formation of Yb_4Bi_3 and YbTe, the present first-principles calculations are reasonable and acceptable.

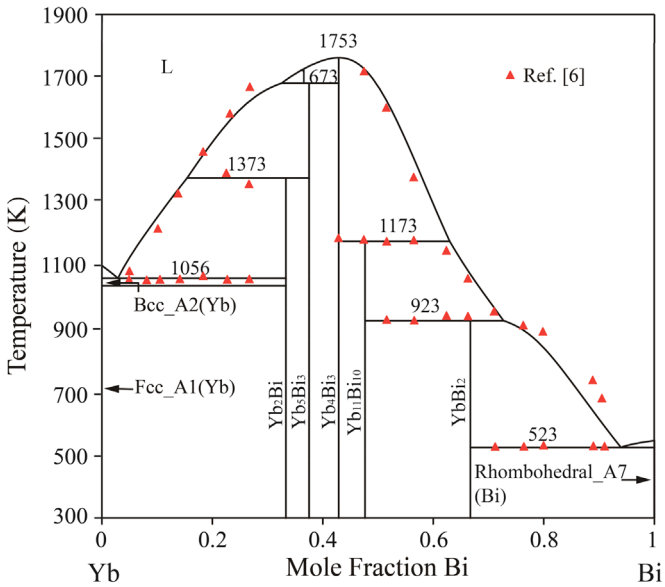

Fig.1. The assessed Yb–Bi phase diagram using the associate model in comparison with the experimental data [6].

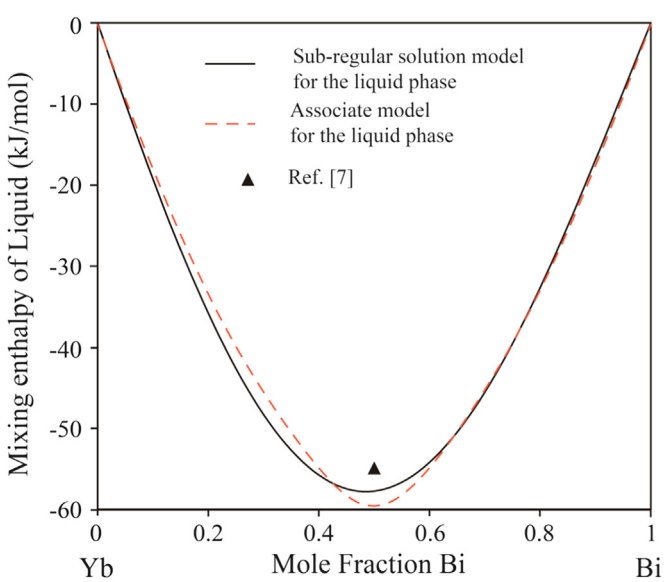

Fig.2. The thermodynamically calculated mixing enthalpy of the liquid Yb–Bi alloy at 2500 K in comparison with the reported value [7]. Reference states are liquid (Yb) and liquid (Bi).

For the compound Yb_5Bi_3, two structures, Yb_5Bi_3-Hexagonal (Yb_5Bi_3-hex for short) and Yb_5Bi_3-Orthorhombic (Yb_5Bi_3-orth for short), are reported in literatures [31,32], and the enthalpies of formation are calculated to be −88574.6 and −87893.8 J/mol, respectively. The calculated results suggest that the Yb_5Bi_3-hex is more stable than Yb_5Bi_3-orth, and is more reasonable to be adopted in the present optimization. For the compounds $Yb_{11}Bi_{10}$ and Yb_2Bi, the enthalpies of formation are not reported in literature and the experimental crystal structure data are not available as well. Therefore, the enthalpies of formation of these two compounds are calculated by the first-principles method using the structures of similar compounds, i.e. the structures of $Yb_{11}Sb_{10}$-Tetragonal [36] and Sm_2Bi-Tetragonal [37] for $Yb_{11}Bi_{10}$ and Yb_2Bi, respectively. These results of the first-principles

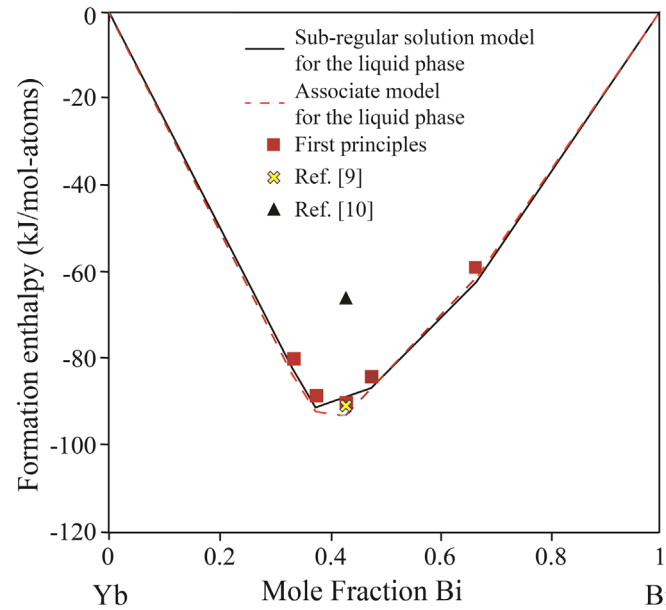

Fig.3. The thermodynamically calculated formation enthalpy of Yb–Bi intermetallic compounds at 300 K using the associate model for the liquid phase in comparison with the present first principles calculation, the experiment result [9] and Miedema model's prediction [10]. Reference states are Fcc_A1 (Yb) and Rhombohedral_A7 (Bi).

calculations are considered in the present assessments to supplement the absence of the thermochemical data of the related compounds.

4.2. Thermodynamic assessment

Based on the lattice stabilities of all pure elements compiled by Dinsdale [27] and the experimental data available from literature, the optimizations of the Yb–Bi/Te binary systems are carried out by means of the Pan Optimizer module of the thermodynamic software Pandat [38].

4.2.1. The Yb–Bi binary system

By using both the associate model and the sub-regular solution model to describe the Gibbs free energy of the liquid phase, two sets of thermodynamic parameters are obtained for the Yb–Bi binary system, as listed in Table 6 in this main text and Table A1 in the attached appendix, respectively.

The calculated phase diagram using the sub-regular solution model is shown in Fig. A1 in the attached appendix. The liquidus boundary is not very well in agreement with that of the experimental data. The calculated phase diagram by using the associate model is shown in Fig. 1, which is nearly identical to the one reported by Borzone [6]. The interaction parameters up to the second order were needed in order to reproduce the satisfactory results by using the associate model. The approximate shape of the liquidus in the Yb–Bi system is determined by interaction parameters of the zero order, ${}^{0}L^{Liq}_{Bi,Yb}$, ${}^{0}L^{Liq}_{Bi,Yb_4Bi_3}$ and ${}^{0}L^{Liq}_{Yb,Yb_4Bi_3}$. The addition of ${}^{1}L^{Liq}_{Bi,Yb_4Bi_3}$ makes the liquidus close to Yb_4Bi_3 in the Yb_4Bi_3-Bi pseudo-binary system move downward, leading to a better agreement with experimental data. ${}^{2}L^{Liq}_{Bi,Yb_4Bi_3}$ is used in order to significantly improve the temperatures of the peritectic reactions $L+Yb_4Bi_3 \leftrightarrow Yb_{11}Bi_{10}$ and $L+Yb_{11}Bi_{10} \leftrightarrow YbBi_2$ for a better agreement with the available experimental data.

The calculated mixing enthalpies of the liquid phase at 2500 K is shown in Fig. 2, together with the results reported by Ref. [7], from which it can be seen that a good agreement is obtained. Fig. 3

Table 7
The thermodynamic parameters of the Yb–Te system by using the associate model for the liquid phase.

Phase	Thermodynamic parameter
Liquid (Te, Yb, YbTe)	$G^{Liq}_{YbTe} = -185936 + 2.84333T + G_YB_LIQ + G_TE_LIQ$
	$G_TE_LIQ = \begin{cases} (298.15 < T < 722.66) \\ +17222.341 - 23.770344*T + GHSERTE - 4.31979 \times 10^{-19}*T^{+7}; \\ (722.66 < T < 1261) \\ +3173.591 + 185.552954*T - 37.656*T*Ln(T) \end{cases}$
	$G_YB_LIQ = \begin{cases} (298.15 < T < 1033) \\ +7030.788 - 40.615571*T - 1.8061816*T*Ln(T) - 0.03250938*T^2 \\ +5.136665 \times 10^{-6}*T^3 - 370554*T^{-1}; \\ (1033 < T < 2000) \\ -6445.835 + 186.690398*T - 36.7774*T*Ln(T). \end{cases}$
	${}^0L^{Liq}_{Te,YbTe} = -51194.8-1.73272T$
	${}^1L^{Liq}_{Te,YbTe} = +20000-15T$
	${}^2L^{Liq}_{Te,YbTe} = -15000$
	${}^0L^{Liq}_{Yb,YbTe} = -20167.84-8.5903T$
	${}^1L^{Liq}_{Yb,YbTe} = 36563-23T$
YbTe	$G^{YbTe}_{Yb,Te} = -155349 + 22.1153T + GHSERYB + GHSERTE$

presents the thermodynamically calculated enthalpies of formation of the intermetallic compounds at 300 K together with the experiment result [9], Miedema model's prediction [10] and the present first-principles calculations. The calculated values by using the present thermodynamic data agree very well with the theoretical results calculated by the present first-principles method and the experimental measurement reported by Ferro et al. [9].

4.2.2. The Yb–Te binary system

The Yb–Te binary system is optimized by using the sub-regular solution model as well as the associate model to describe the Gibbs free energy of the liquid phase. One set of thermodynamic parameters is obtained for the Yb–Te binary system by using the sub-regular solution model, the result of which is almost the same as the one reported by Ghamri et al. [12]. The other set is obtained by using the associate model, as listed in Table 7. The calculated phase diagram of the Yb–Te system by using the associate model is shown in Fig. 4, together with the experimental data from Abrikosov et al. [11] and the optimization result by Ghamri et al. [12] using sub-regular solution model. All the calculated invariant reactions in the Yb–Te system are listed in Table 2. From Fig. 4 and Table 2 it can be seen that by using the associate model the calculated results are in a much better agreement with most of the experimental data [11] than by using sub-regular solution model.

The calculated enthalpy of formation of the only intermetallic compound YbTe at 298 K is shown in Fig. 5, indicating the fairly good agreement with the experimental data from Ref. [13] and a relative deviation of 6.5 % from the present first-principles prediction.

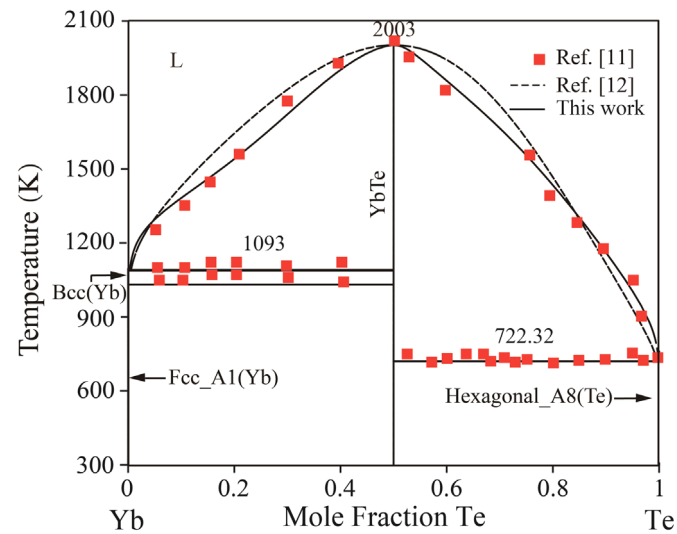

Fig.4. The assessed Yb–Te phase diagram in comparison with the experimental data [11] and the optimization result by Ghamri et al. [12] using sub-regular solution model.

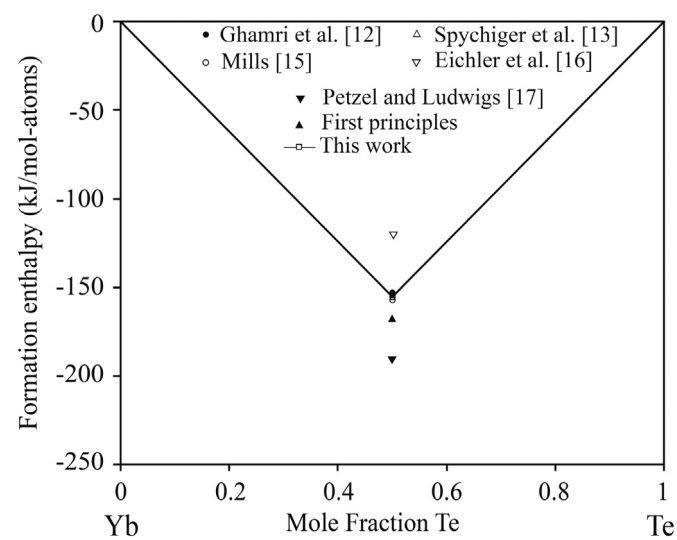

Fig.5. The thermodynamically calculated formation enthalpy of Yb–Te intermetallic compound at 298 K in comparison with the present first principles calculation and the reported value [12,13,15–17]. Reference states are Fcc_A1 (Yb) and Hexagonal_A8 (Te).

5. Conclusions

With the assistance of the present first-principles calculations and the experimental data of the phase diagrams and the thermochemical properties available in literatures, the Yb–Bi/Te binary systems are thermodynamically assessed. In consideration of the asymmetric shapes of the liquidus around the compounds Yb_4Bi_3/YbTe in the Yb–Bi/Te phase diagrams, not only the sub-regular solution model but also the associate model are used to describe the Gibbs free energies of the liquid phases for both of the systems. The consistent sets of the thermodynamic parameters for the descriptions of various phases of these two systems have been obtained. By using the associate model for the Yb–Bi/Te systems, all of the calculated temperatures and compositions of the

Table A1
The thermodynamic parameters of the Yb—Bi system by using the sub-regular solution model for the liquid phase.

Phase	Thermodynamic Parameter
Liquid (Bi, Yb)	${}^{0}L^{Liq}_{Bi,Yb} = -231036-1.9716T$
	${}^{1}L^{Liq}_{Bi,Yb} = +16445.2-20T$
	${}^{2}L^{Liq}_{Bi,Yb} = +47014+4.8599T$
Yb_2Bi	$G^{Yb_2Bi}_{Yb:Bi} = -82092.9+8.85816T+0.6667GHSERYB+0.3333GHSERBI$
Yb_5Bi_3	$G^{Yb_5Bi_3}_{Yb:Bi} = -91117.2+9.68389T+0.6250GHSERYB+0.3750GHSERBI$
Yb_4Bi_3	$G^{Yb_4Bi_3}_{Yb:Bi} = -88724.2+4.4362T+0.5714GHSERYB+0.4286GHSERBI$
$Yb_{11}Bi_{10}$	$G^{Yb_{11}Bi_{10}}_{Yb:Bi} = -86621+5.56052T+0.5238GHSERYB+0.4762GHSERBI$
$YbBi_2$	$G^{YbBi_2}_{Yb:Bi} = -62244.5+1.60568T+0.3333GHSERYB+0.6667GHSERBI$

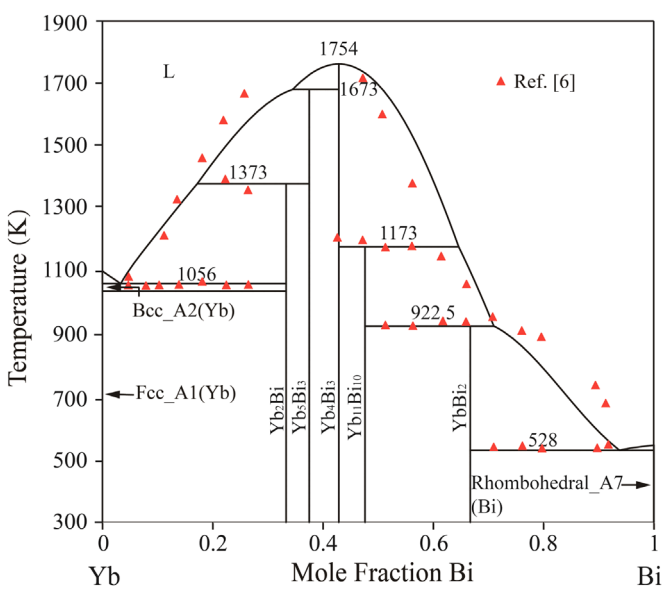

Fig.A1. The assessed Yb–Bi phase diagram using the sub-regular solution model in comparison with the experimental data [6].

invariant reactions as well as the thermochemical properties are in good agreement with the experimental data. Especially, the calculated liquidus boundaries of both systems are improved substantially.

When ternary or multi-component systems are considered, the description by using the sub-regular solution model is easy to use technically. Whereas the associate model is worthy of trying after figuring out the existing associated species in the liquid phase of the system, so that the phase equilibrium relations and the thermochemical properties of the system can be expressed properly.

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Appendix A

The thermodynamic parameters optimized by using the sub-regular solution model to describe the Gibbs free energy of the liquid phase for the Yb–Bi binary system are listed in Table A1. The calculated phase diagram is shown in Fig. A1 together with the experimental data available from literature for comparison.

Appendix A. Supplementary material

Supplementary data associated with this article can be found in the online version at http://dx.doi.org/10.1016/j.calphad.2015.10.007.

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